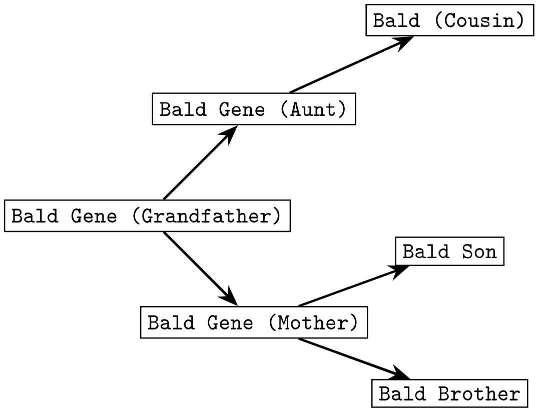
Q1.



In the causal system above with binary variables, `Bald Son` and `Bald Brother` are associated because they share a common cause (`Baldness Gene`). If we condition on `Baldness Gene (Mother)`, what happens to their association?

A. They become independent.

B. They become dependent.

C. It depends on whether they are twins.

D. They remain associated, but their dependence on `Bald Cousin` is eliminated.

Q2. A researcher analyzes the distribution of salaries for a large company and finds it is strongly skewed to the right. Which pair of numerical summaries would be most appropriate and reliable for describing the center and spread of this salary data?

A. Mean and Standard Deviation.

B. Mean and Range.

C. Mode and Range.

D. Median and Interquartile Range (IQR)

Q3. Several pieces of fruit from each tree in an orchard are selected. What sampling technique is being used in this scenario?

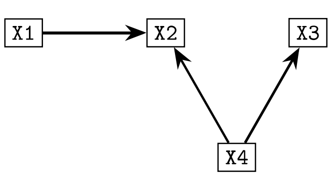
A. stratified sampling.

B. systematic sampling.

C. simple random sampling.

D. cluster sampling.

Q4. For the diagram below, which option correctly describes the system after an ideal intervention on `X2`?



A. A graph where intervention arrows point to `X1` and `X4`, breaking any arrows that might point into them.

B. A graph where the intervention arrow points to `X2`, but all original arrows remain intact.

C. A graph where the intervention arrow points to `X2`, and the arrow from `X2` to `X3` is removed.

D. A graph where the intervention arrow points to `X2`, and the arrows from `X1` to `X2` and `X4` to `X2` are removed.

Q5. A survey taken in a large statistics class contained the question: "What's the fastest you have driven a car (in miles per hour)?" The five-number summary for the 87 males surveyed is:

min = 55, Q1 = 95, Median = 110, Q3 = 120, Max = 155.

Should any observation in this data set be classified as an outlier?

A. We need to know $\bar{x}$ and n to decide.

B. Yes.

C. We need to know the population parameters to decide.

D. No.

Q6. How would an 'instrumentalist' view a highly successful scientific theory, such as the theory of atoms?

A. As a useful intellectual tool for organizing observations and making predictions, but not as a literally true or false description of reality.

B. As a literally true description of the unobservable world, because its success would be a miracle otherwise.

C. As a claim that is in fact either true or false, but one we should remain agnostic about due to the limits of justification.

D. As a factual statement that has graduated from being a hypothesis to being a law of nature.

Q7. Which of the following best describes the relationship between the concepts of 'theory', 'hypothesis', and 'law'?

A. They describe different, non-exclusive features of a scientific claim:its subject matter, its justification status, and its generality.

B. They are mutually exclusive; a claim can be a theory, a hypothesis, or a law, but more than one at a time.

C. A claim progresses from a hypothesis to a theory and finally graduates to the status of a law once it is proven.

D. A law is a mathematical theory, while a hypothesis is a non-mathematical theory.

Q8. Which of the following is an example of an 'internal virtue' of a theory?

A. The theory is compatible with other well-established scientific beliefs.

B. The theory accurately predicts the outcome of a new experiment.

C. The theory leads to the development of a useful new technology.

D. The theory successfully explains a puzzling natural phenomenon.

Q9. In the 'Big Picture of Statistics', what is the primary role of Probability?

A. To ensure the sample perfectly represents the entire population.

B. To allow for drawing conclusions about a population based on sample data.

C. To summarize the collected sample data into graphs and tables.

D. To identify the entire group that is the target of our interest.

Q10. What is the central problem of ‘underdetermination’ in the philosophy of science?

A. Scientists often lack the imagination to come up with enough creative theories to explain all the available data.

B. Experiments are always too complex and expensive to provide conclusive proof for any single theory.

C. The available observational evidence is not sufficient to uniquely determine which of several competing, empirically equivalent theories is true.

D. A theory cannot be considered scientific until it has been determined to be true beyond all possible doubt.

Q11. If a political poll of 1,650 adults has a margin of sampling error of ± 3 percentage points, what would happen to the margin of error if the pollsters only surveyed a subgroup of 455 of those adults?

A. The margin of error would increase because the sample size is smaller.

B. The margin of error would stay the same because they are part of the original sample.

C. The margin of error would become zero because the subgroup is not a random sample.

D. The margin of error would decrease because the group is more specific.

Q12. A major challenge for scientists who seek the true account of the world is underdetermination, where multiple theories can achieve a perfect score on external virtues. In this situation, what is the primary role of internal virtues?

A. To limit the collection of empirically equivalent theories by offering criteria – such as preferring the simplest or most coherent account – to justify the inference that one theory is more likely to be true.

B. To demonstrate the necessity of adopting an instrumentalist view, where theories are only judged by their usefulness, not their truth.

C. To provide definitive, deductive proof that a theory corresponds to reality by linking its structure directly to observable phenomena.

D. To eliminate the need for any further testing, as external confirmation is deemed unreliable by realists.

Q13. The annual salary of teachers in a certain state X has a mean of $55,000 and standard deviation of σ = $6,000.

What is the probability that the mean annual salary of a random sample of 9 teachers from this state is more than $59,000?

A. 2.5%.

B. 22%.

C. We do not have enough information to solve the problem.

D. 16%.

Q14. Two researchers analyze digital surveillance policies. One focuses on how technologies improve efficiency (“rational lens”), the other examines how surveillance reinforces political power (“political lens”). Which of the following insight best reflects the research process?

A. Both are positivist because they analyze organizations

B. Different paradigms emphasize different aspects of the same reality

C. Only the rational lens is valid because it uses objective data

D. These are competing opinions, not research

Q15. You overhear this argument: “This is the second time you’ve caught her cheating; therefore, you must flunk her.” Which option best analyzes the reasoning?

A. It is clearly inductive because it increases the likelihood of flunking.

B. It is deductive, because the context strongly suggests an unstated rule that any student caught cheating twice should be flunked.

C. It is neither inductive nor deductive because it lacks any universal statement.

D. It is inductive unless the speaker explicitly states the universal rule.

Q16. In order to estimate μ, the mean weight of a typical passenger + his/her luggage on an early morning flight from New York City to Washington DC, an airline company chose a random sample of 16 passengers who took the flight. From past research on similar flights, it is known that weights of passengers + their luggage have a standard deviation σ = 20 pounds.

Can we safely estimate μ with the confidence interval that we developed?

A. Sometimes.

B. Yes, by using the Central Limit Theorem.

C. Yes, by using the Law of Large Numbers.

D. No.

Q17. In the realistic model of hypothetico-deductive testing, the apparent deductive certainty of falsification is lost when auxiliary theories are included. Why does a failed prediction not unambiguously falsify the hypothesis in this scenario?

A. The hypothesis is always more entrenched than the auxiliary theories, granting it immunity from falsification unless all auxiliaries have been deductively proved true.

B. The only factors that can absorb the blame for a failed prediction are the experimental conditions, which are directly checkable and correctable.

C. The deduction relies on the truth of the entire group (hypothesis plus all auxiliaries), meaning failure only proves that not all members are true, forcing scientists to rely on internal virtues like entrenchment to decide where to assign blame.